CAPSTONE: END-TO-END ANALYTICS

Data Integration, Cleaning, Transformation & Business Dashboard

Week 4 — Task 4.4

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Project report documenting the analytics workflow, data quality checks, dashboard design, findings, limitations, and recommendations.

Executive Summary

This capstone documents an end-to-end analytics workflow that integrates two source datasets, inspects data quality, applies a reusable cleaning function, exports an analysis-ready CSV, and uses that dataset as the basis for a business dashboard. The supplied notebook uses pandas and NumPy, performs a cross join because the source tables do not share a common key, checks missing and duplicate records, standardizes column names, converts customer_id to a string when present, and exports the cleaned result. The supplied dashboard dataset contains 312 rows and 11 columns. Across the dataset, total sales are 166,140,000 FCFA and total quantity is 598 units. The report also describes dashboard visuals and filters that support exploration by product, region, payment method, sales representative, and insurance status.

Project Objectives

- Integrate the source data into a single analysis-ready dataset.
- Inspect dataset structure and identify missing or duplicate records.
- Build a reusable cleaning function to standardize columns and handle quality issues.
- Export the cleaned dataset for downstream analysis and dashboarding.
- Summarize business performance through KPIs, charts, and actionable findings.

1. Introduction

Organizations often collect information in separate files. Before meaningful analysis can happen, these sources need to be loaded, inspected, combined appropriately, cleaned, and prepared for reporting. This project demonstrates a practical analytics pipeline using Python and a dashboard concept for communicating the results to business users.

Problem Statement

Separate source datasets are difficult to analyze together consistently. The project establishes a repeatable workflow that brings data into one table, checks data quality, applies standardized cleaning steps, and makes the resulting data available for business reporting.

Scope

The scope includes loading the insurance and sales CSV files, integrating them, inspecting the combined data, cleaning the data, exporting the cleaned CSV, summarizing business KPIs, and defining a dashboard layout.

2. Tools and Technologies

Tool	Purpose
Python	Runs the data preparation and analysis logic.
pandas	Loads CSV files, combines tables, inspects quality, transforms columr and exports results.
NumPy	Imported for numerical operations and compatibility with the analysis workflow.
Jupyter Notebook	Provides an interactive, documented environment for executing the pipeline.
Power BI	Used to present KPIs, charts, and slicers in an interactive business dashboard.

3. Data Integration and Pipeline Design

The notebook loads two CSV files: insurance_data.csv and sales_dataset.csv. It previews each source with head(), then combines them using pandas merge(..., how='cross'). A cross join pairs every row from the first table with every row from the second table. This approach is appropriate only when every combination is intentionally required; it can greatly increase row counts and does not establish a real-world customer-to-sale relationship by itself.

Pipeline Stages:

Stage	Description
1. Import libraries	Import pandas and NumPy.
2. Load sources	Read the insurance and sales CSV files.
3. Preview source data	Use head() to inspect the first records from each source.
4. Integrate	Use a cross join because the notebook states there are no common keys.
5. Inspect	Check shape, data types, missing values, and duplicate rows.
6. Clean	Standardize column names; convert customer_id to string if present; remove duplicates and rows with missing values.
7. Export	Write the cleaned dataset to cleaned_integration_data.csv.

4. Data Quality and Transformation

The notebook checks the combined data before cleaning. The shape and info() outputs describe the size and types of the dataset, while isnull().sum() and duplicated().sum() quantify missing values and duplicate rows. The reusable clean_pipeline() function copies the input, normalizes column names to lowercase with underscores, converts customer_id to text when available, removes duplicates, and drops rows containing missing values.

Current cleaned dataset checks:

Check	Observed result
Rows	312
Columns	11
Missing cells remaining	0
Duplicate rows remaining	0

Important data-quality note: dropping every row with any missing value can remove useful information. In a production workflow, the missingness should be examined column by column and a justified fill/drop rule chosen. Also, a cross join should be confirmed as a business requirement; if the sources have a shared identifier, a keyed join is generally preferable.

5. Business Analysis and Dashboard Design

The dashboard is intended to summarize sales performance and customer insurance status from the cleaned integration dataset. The KPI cards and visuals below should be built in Power BI after importing the cleaned CSV.
Slicers make the report interactive, allowing a viewer to filter all applicable visuals by selected categories.

Key Performance Indicators ;

KPI	Value	Interpretation
Total Sales	166,140,000 FCFA	Sum of total_sales across records.
KPI	**Value**	**Interpretation**
Total Quantity	598	Total units represented in quantity.
Unique Customers	12	Distinct customer_id values in the dataset.
Insurance Rate	53.85%	Share of records with bought_insurance = 1.

Recommended Dashboard Visuals ;

Visual	Fields / setup	Purpose
Card	Sum of total_sales	Overall sales performance.
Card	Sum of quantity	Total units sold.
Card	Sum of unit_price	Total of unit price sold
Card	Insurance Rate measure	Share of insured records.
Bar chart	product; Sum of total_sales	Compare sales by product.
Line chart	region; Sum of total_sales	Compare regional performance.
Slicer dropdown	sales_rep	Shows sales_rep data
Input slicer	region	Show specific region data
Pie chart	bought_insurance; age	Compare insurance status.
Slicer	Payment method	Filter

6. Key Findings

The highest-revenue product is Laptop with 70,200,000 FCFA. The lowest-revenue product is Tablet with 37,440,000 FCFA.

The highest-sales region is Buea (89,440,000 FCFA), while the lowest-sales region is Limbe (29,900,000 FCFA).

The leading sales representative by recorded revenue is Sona (59,800,000 FCFA).

The leading payment method by recorded revenue is Mobile Money (67,860,000 FCFA).

These findings describe the supplied dataset and should not be interpreted as causal explanations. Business context, time coverage, and the implications of the cross join should be confirmed before making operational decisions.

7. Limitations and Risks

- The report is based on the cleaned dataset available for this task; the original source CSV files were not independently revalidated in this report.
- A cross join can create many combinations that do not represent real transactions or actual customer relationships.
- Dropping all rows containing missing values may introduce bias if missingness is systematic.
- The insurance-rate KPI is calculated as a share of records, not necessarily a share of unique customers; if the dataset repeats customers, the distinction matters.

- The report describes a Power BI dashboard design and configuration but does not claim that a native .pbix filewas generated.

8. Recommendations and Next Steps

- Validate whether a cross join is analytically intended; use a shared key for a relational merge if one exists.
- Document row counts before and after each transformation to make the pipeline auditable.
- Review missing values by column before deciding whether to drop or impute records.
- Add data validation checks for numeric fields, categories, and sales calculations.
- Build and save the interactive report in Power BI Desktop as a .pbix file, then export PDF or screenshots for evidence.
- Add a README with setup instructions, data dictionary, pipeline steps, dashboard screenshots, and limitations.

9. Conclusion

This capstone demonstrates the main stages of an end-to-end analytics workflow: source loading, data integration, quality inspection, reusable cleaning, export, and business reporting design. The Python notebook provides a repeatable foundation, while the proposed Power BI dashboard translates the cleaned dataset into performance indicators and interactive comparisons. The most important improvement before relying on the results is to validate the cross-join logic and ensure that the resulting rows represent the intended business entities.

Appendix: Dataset profile

Dataset file used for summary: cleaned_integration_pipeline_data.csv. Profile: 312 rows × 11 columns. Missing cells: 0. Duplicate rows: 0.

The notebook exports a file named cleaned_integration_data.csv; the file used here is the cleaned integration dataset supplied in the working session.